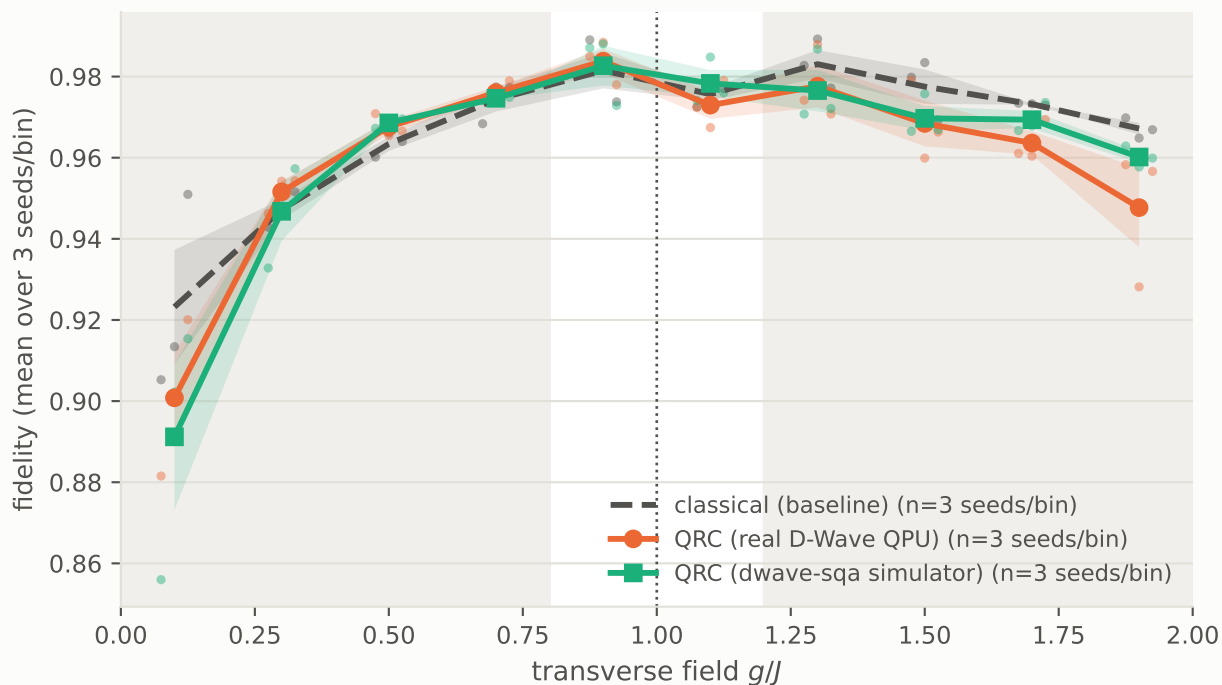
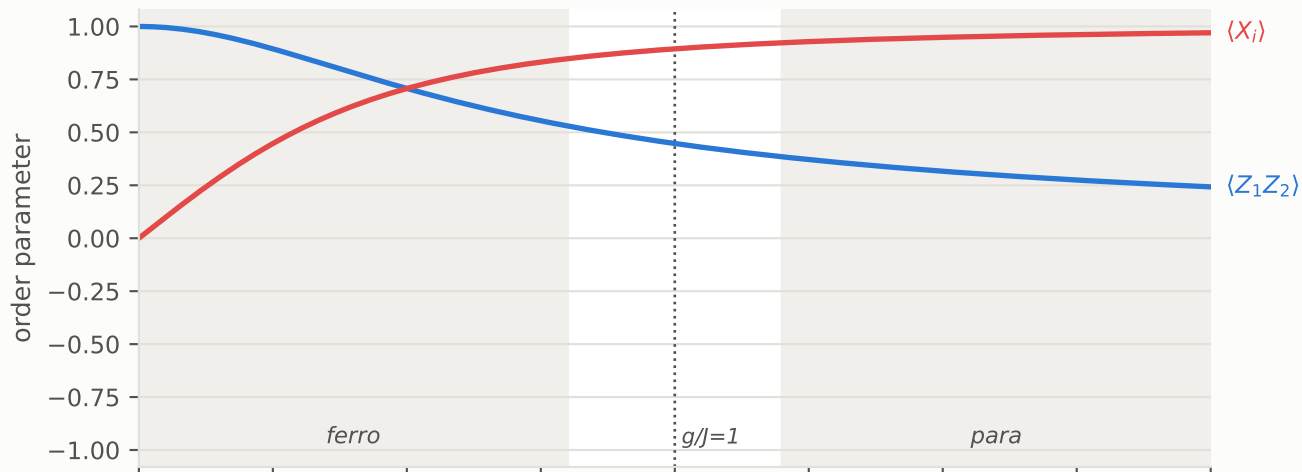


TFIM ground state (N=2): physical phase diagram vs.
D-Wave reconstruction fidelity (real QPU vs. simulator)



Real D-Wave QPU (Advantage_system4) vs. dwave-sqa simulator, matched settings: $n_{\text{train}}=n_{\text{val}}=100$, $n_{\text{test}}=200$, 3 seeds, num_reads=20, 10 g-bins. Shaded bands: ferro/para ranges used elsewhere in this document. random control omitted.